

## Related Works: Research Papers on PF-MARL for Epidemic Control

| S.No | Title                                                                                                                            | Category  | Key Focus / Relevance                                                                                                    |
|------|----------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------|
| 1    | Planning Multiple Epidemic Interventions with Reinforcement Learning                                                             | Core MARL | Strategizing sequential and concurrent public health interventions using RL.                                             |
| 2    | Applying Reinforcement Learning to Epidemic Management: Strategic Influenza Control in Multiple Scenarios                        | Core MARL | Formulating dynamic RL-based strategies specifically designed for controlling influenza spread across diverse scenarios. |
| 3    | H2-MARL: Multi-Agent Reinforcement Learning for Pareto Optimality in Hospital Capacity Strain and Human Mobility during Epidemic | Core MARL | Achieving a Pareto-optimal balance between reducing hospital strain and preserving human mobility.                       |
| 4    | HRL4EC: Hierarchical Reinforcement Learning for Multi-Mode Epidemic Control                                                      | Core MARL | Using a multi-level RL approach to manage complex, multi-tiered epidemic interventions.                                  |
| 5    | Deep Reinforcement Learning for Large-Scale Epidemic Control                                                                     | Core MARL | Applying deep RL frameworks to handle continuous, large-scale agent environments for infection control.                  |
| 6    | DPEpiNN: A Unified Framework for Differentially Private Epidemic Forecasting                                                     | DP        | Combining deep neural networks with compartmental models under strong differential privacy bounds for forecasting.       |
| 7    | Epidemiological Modeling with Mobile Phone Data under Differential Privacy                                                       | DP        | Safe utilization of real-world human mobility data using DP to preserve individual anonymity.                            |
| 8    | Controlling Epidemic Spread on Networks under Edge Differential Privacy                                                          | DP        | Protecting sensitive patient contact-network graphs while optimizing vaccination strategies.                             |
| 9    | Differentially Private Local Distributed Reproduction Numbers                                                                    | DP        | A framework to securely calculate localized R0 values without leaking node-to-node interaction data.                     |
| 10   | EXAM: A Federated Learning System for Predicting Future Oxygen Requirements of COVID-19 Patients                                 | FL        | Large-scale 20-institute collaborative FL framework validating improved predictive generalizability.                     |
| 11   | Federated Deep Learning for Predicting Mortality of COVID-19 Patients                                                            | FL        | Deploying deep learning across distributed hospital datasets to predict clinical outcomes without sharing raw PI.        |

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| 12 | FedPandemic: A Cross-Device Federated Learning Framework for Epidemic Prognosis                          | FL        | Integrating cross-device learning and DP-related noise algorithms to assess symptom prognosis securely.                      |
| 13 | A Multi-Agent Reinforcement Learning Framework for Public Health Decision Analysis                       | Core MARL | Employing deep MARL for cross-jurisdictional decision modeling and balancing localized resource limits.                      |
| 14 | Cognitively-plausible reinforcement learning in epidemiological agent-based simulations                  | Core MARL | Extending RL inside agent-based models to balance an individual's economic incentives against community health restrictions. |
| 15 | A Reinforcement Learning Based Decision Support Tool for Epidemic Control: Validation Study for COVID-19 | Core MARL | Validating automated RL support tools for recommending policies that minimize socioeconomic costs.                           |